

Marketing Applications, Inc.: Migrate to Virtual Machines and support from PSO are two important ingredients for a successful migration

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ABOUT MARKETING

 Marketing Applications

About Marketing Applications

Marketing Applications is a holding company that contains three companies: Data Technology Lab, Inc., Marketchi, Inc., and Programming Fast Inc. Marketing Applications provides platform services for the marketing research industry sector; it provides service to more than 200 companies, including market research companies, advertising agencies, manufacturers, think tanks, consulting firms, and a variety of media companies. Currently Marketing

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solutions in the n
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We interviewed th
company's chief
officer and techn
about the project

Adopting Google Cloud
of its simple interface
extensive feature set.
"live migration" of VM

Marketing Applications, an IT solution in the
marketing research area, provides a range of
survey systems, aggregation tools, consumer panels
and operational resources that support researchers
who need advanced analysis and insights for
companies such as research companies, consumer
goods manufacturers, and advertising agencies.
They support marketing research in terms of both
technology and human resources.

Applications' market share in Japan is number
one.

Industries: Technology

Location: Japan

(<https://cloud.google.com/kubernetes-engine/>),
Migrate to Virtual Machines
(<https://cloud.google.com/migrate/virtual-machines>)

Compute Engine
(<https://cloud.google.com/compute/>)

Cloud Storage (<https://cloud.google.com/storage/>)
(<https://cloud.google.com/products/operations>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Virtual Private Cloud
(<https://cloud.google.com/vpc/>)

Cloud Interconnect
(<https://cloud.google.com/interconnect/>)

Cloud VPN
(<https://cloud.google.com/hybrid-connectivity/>)

Operations
(<https://cloud.google.com/products/operations>)

The following four services are mainly provided by the company;

- MApps for Survey that support creation, distribution, and collection of survey screens
- MApps for Analysis that support collecting survey results and creating analysis reports
- MApps Panel, a research platform that provides a large-scale market-research-dedicated panel targeting more than 2 million people
- Surveyoid, a self-administered survey tool

In the research platform, Marketing Applications used a data center to build a virtual infrastructure in an on-premises environment where servers, networks, and other equipment were installed for operational management. But maintaining and managing an on-premises environment had become difficult due to issues such as cost, operation, and technology. So they decided to first lift their on-premises environment to the public cloud and eventually shift to cloud native.

Lead engineer Takashi Toyobe says, “Due to the deterioration of the on-premises environment, problems occurred in the virtual infrastructure on a daily basis, and it took a long time to recover, affecting service. Expanding capacity was also an issue, and it was not flexible to procure servers and network equipment, install them in the data center, and perform various settings. Adding to that, maintenance of the on-premises environment was running out.”

Google Cloud (<https://cloud.google.com/>) was selected as the infrastructure provider to solve these issues

Marketing Applications was facing.

Hajime Igari, Executive Officer and CTO, explains the reason for adopting Google Cloud as follows; “The biggest challenge was how to reduce the increasing operational load. The biggest reason we chose Google Cloud was that we thought it would be simple and have sufficient functions, and that live migration would make management easier. In particular, we feel that live migration is fast and very good.”



Mr. Hajime Igari, Operating Officer & CTO, Marketing Applications

Easy management with containerization

Lifting and shifting the on-premises environment took place in three phases. Phase 1 of the April–June 2019 period completed the lift and shift of MApps for Survey, the main service of the research platform. First, Marketing Applications shifted the

functions for clients on MApps for Survey to Google Kubernetes Engine.

Since MApps for Survey is a multi-tenant product, the company needs to be able to flexibly provide a web server according to the service plan it provides to clients. Therefore, optimal resource allocation is required and containerization is effective.

"Previously, if we needed additional resources, we had to manually scale out by creating and configuring an instance. With Kubernetes, we can automatically scale out within the upper limit, depending on the operation status. Once the initial maintenance is done, the operation management will be much easier after that. Thus we can establish a correct operation system," says Mr. Igari.

By leveraging Migrate to Virtual Machines (<https://cloud.google.com/migrate/virtual-machines>), Marketing Applications has lifted its server functions to Google Cloud. In Phase 2, the company migrated products provided to clients other than MApps for Survey, and in Phase 3 it migrated server functions other than MApps for Survey. Phase 1 and Phase 2 completed in November 2019. Phase 3 completed in May 2020.

When lifting Research Platform, Marketing Applications built the basic server functionality with an instance of Compute Engine. For the shift, it adopted Google Cloud Platform (<https://cloud.google.com/>) (GKE) for containerization of MApps for Survey. Cloud Storage (/storage) and

Filestore (/filestore) are also used as storage environments.

Mr. Toyobe talks about the effects of adopting Google Cloud as follows. "Compute Engine gives you the flexibility to optimize costs because you don't have to buy any resources in advance. You automatically get sustained-use discounts when you use it, and you won't be charged when you're not using it. Also, in an on-premises environment, we had to ask infrastructure experts to reconfigure the infrastructure when necessary, but Google Cloud is very convenient because application developers can easily reconfigure the infrastructure. Although containerization specialists are required as containerization progresses, GKE has also made container management easier."



Mr. Takashi Toyobe, Lead Engineer, Marketing Applications

**Support from Google PSO
consulting services, enabling**

quick migration with a small number of members

Migrate to Virtual Machines was used for migration of the on-premises environment, but it was a little tricky because there were old operating systems that did not support Migrate to Virtual Machines. So Marketing Applications decided to consult with the Google Professional Services Organization (PSO).

Mr. Igari looks back on those days: "The suggestion to try PSO's support was part of a regular monthly meeting that started around April 2019. We understood that it is necessary to proceed with making decisions based on the cases where the tools can be used for the migration as is, and the cases where it is better to change the architecture for the migration, because the migration depends on the environment. We decided to use the consulting service to make the right decision without any rework and to proceed with the migration efficiently."

For example, when lifting a mail delivery system sends mail from multiple global IP addresses to Google Cloud, PSO proposed a method that allows sending from multiple global IP addresses by setting a load balancer and forwarding rules. Marketing Applications accepted and adopted this. If a problem occurred during the process, it was of great value that the PSO could immediately judge whether it was a malfunction or a design error.

"Thanks to the PSO team's swift judgment, we were able to achieve this scale of migration with a small number of people and in a short period of about two

months. At first we were trying to complete it ourselves, but now we think it was reckless. If there are similar projects in the future, we would like to use consulting services," says Mr. Toyobe.

"Having a Kubernetes cluster running on GKE is a big plus. This eliminates the anxiety about capacity. This is an important point from the perspective of operating the service in the long term. Currently, not all requirements have been met, but we think it was a very satisfying project as a preliminary preparation for shifting to cloud native in the future," adds Mr. Igari.

